

SET 1 : MONTHLY SALES DATA

1.

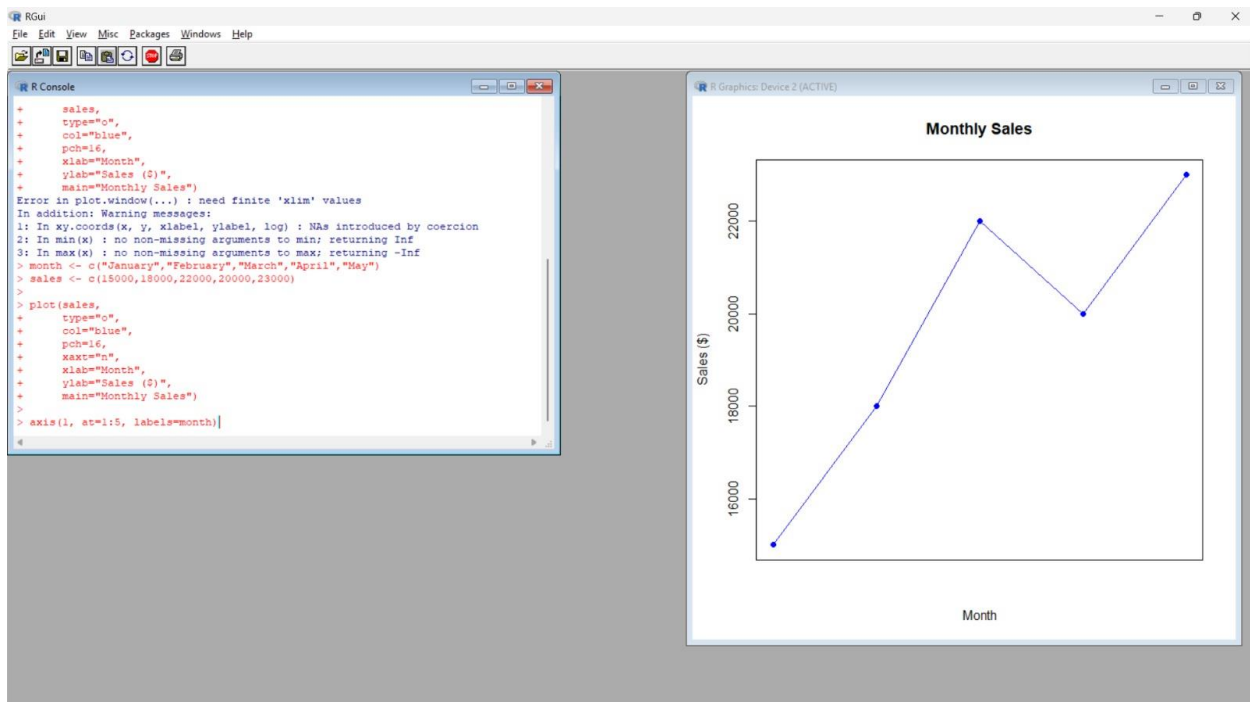
CODE :

```
library(ggplot2)
```

```
sales_data <- data.frame(  
  Month = factor(c("January", "February", "March", "April", "May"),  
    levels = c("January", "February", "March", "April", "May")),  
  Sales = c(15000, 18000, 22000, 20000, 23000)  
)
```

```
ggplot(sales_data, aes(x = Month, y = Sales, group = 1)) +  
  geom_line(color = "blue", linewidth = 1.2) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Monthly Sales",  
    x = "Month",  
    y = "Sales ($)"  
  ) +  
  theme_minimal()
```

OUTPUT :



2.

CODE :

```
library(ggplot2)
```

```
product_data <- data.frame(  
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),  
  Sales = c(45000, 38000, 52000, 41000, 47000)  
)
```

```
ggplot(product_data, aes(x = Product, y = Sales, fill = Product)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Top-Selling Products",  
    x = "Products",  
    y = "Annual Sales ($)"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```

OUTPUT :



3.

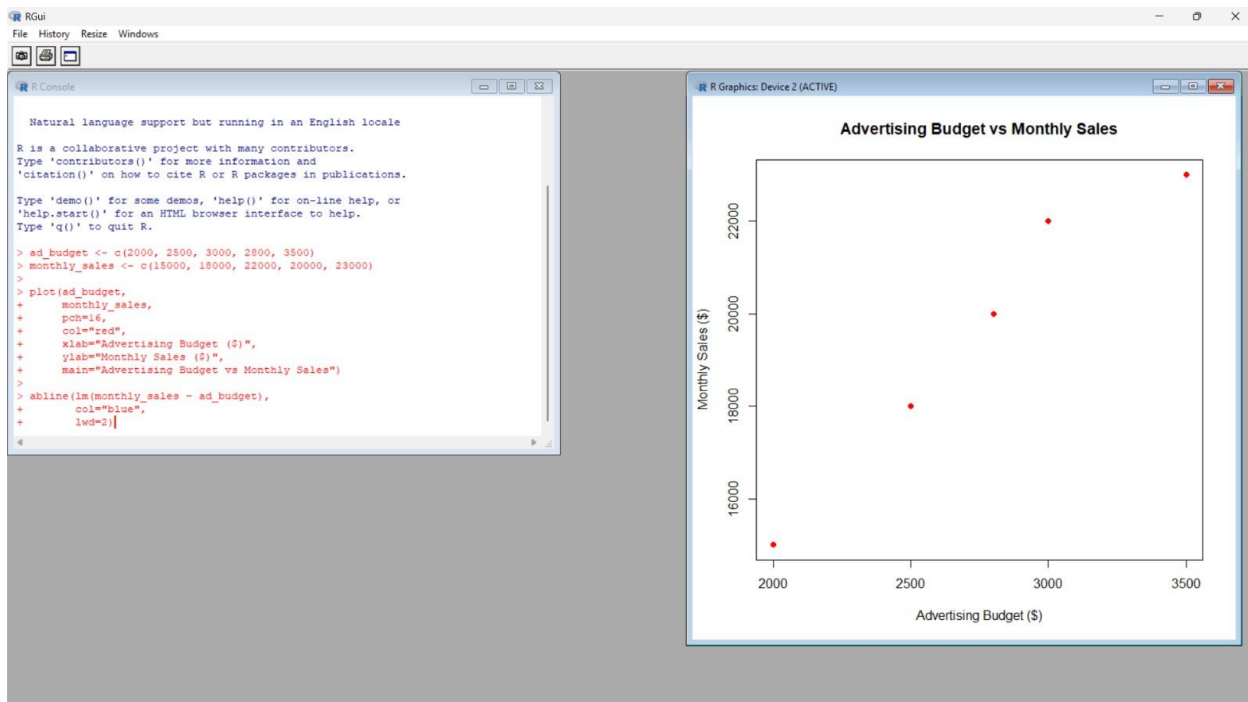
CODE:

```
library(ggplot2)
```

```
sales_data <- data.frame(  
  Advertising_Budget = c(2000, 2500, 3000, 2800, 3500),  
  Monthly_Sales = c(15000, 18000, 22000, 20000, 23000)  
)
```

```
ggplot(sales_data, aes(x = Advertising_Budget, y = Monthly_Sales)) +  
  geom_point(color = "blue", size = 4) +  
  labs(  
    title = "Advertising Budget vs Monthly Sales",  
    x = "Advertising Budget ($)",  
    y = "Monthly Sales ($)"  
  ) +  
  theme_minimal()
```

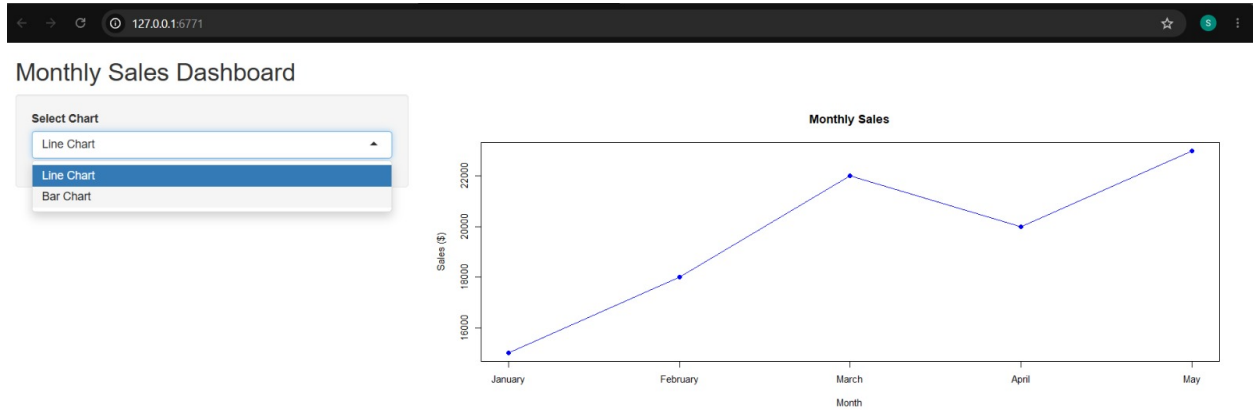
OUTPUT :



4.

CODE :

OUTPUT :



SET 2 : CUSTOMER FEEDBACK ANALYSIS

1:

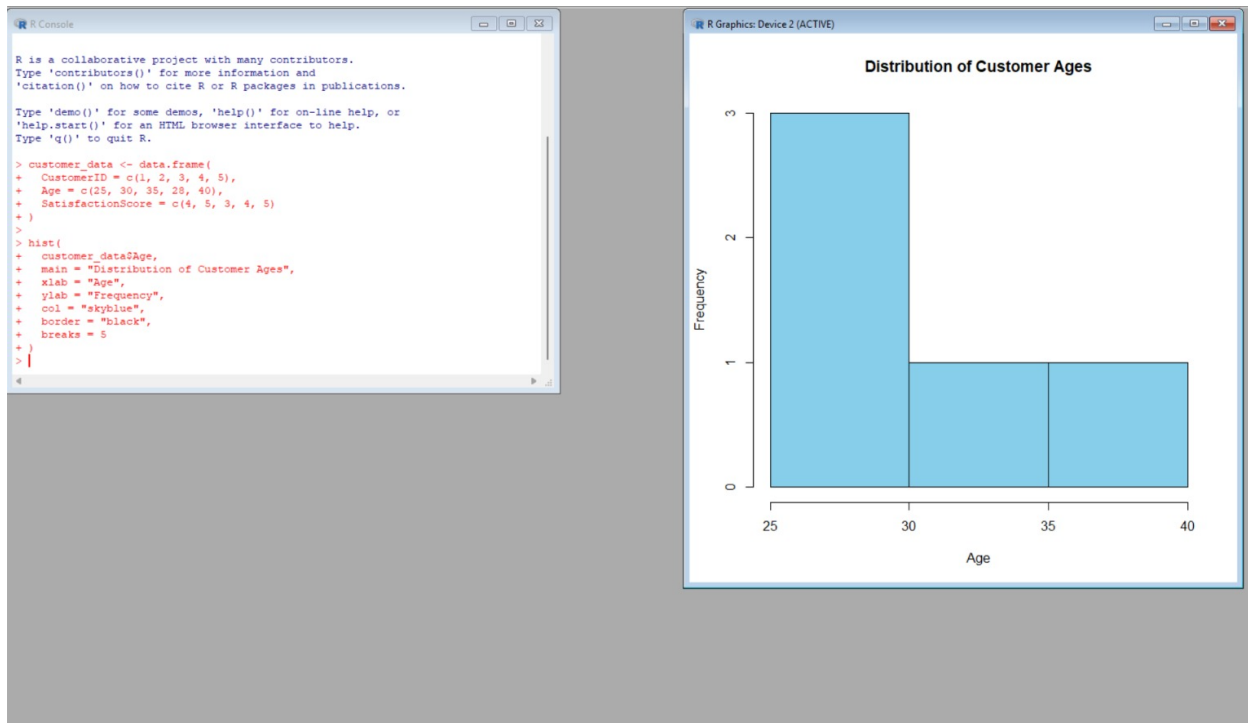
CODE :

```
library(ggplot2)
```

```
customer_data <- data.frame(  
  Customer_ID = c(1, 2, 3, 4, 5),  
  Age = c(25, 30, 35, 28, 40),  
  Satisfaction_Score = c(4, 5, 3, 4, 5)  
)
```

```
ggplot(customer_data, aes(x = Age)) +  
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +  
  labs(  
    title = "Distribution of Customer Ages",  
    x = "Age",  
    y = "Frequency"  
  ) +  
  theme_minimal()
```

OUTPUT :



2 :

CODE :

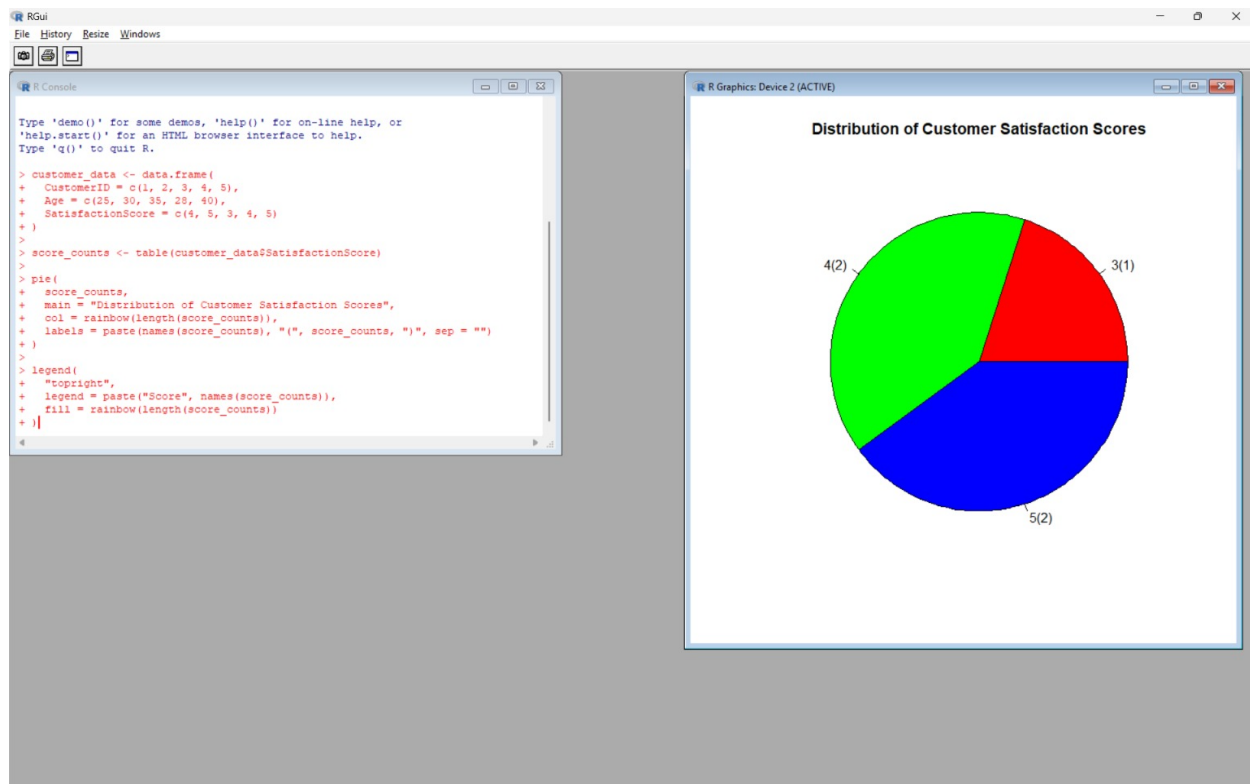
```
library(ggplot2)
```

```
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  Satisfaction_Score = c(4, 5, 3, 4, 5)
)
```

```
score_data <- as.data.frame(table(customer_data$Satisfaction_Score))
colnames(score_data) <- c("Score", "Count")
```

```
ggplot(score_data, aes(x = "", y = Count, fill = Score)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  labs(
    title = "Distribution of Customer Satisfaction Scores",
    fill = "Score"
  ) +
  theme_void()
```

OUTPUT :



3 :

CODE :

```
library(ggplot2)
```

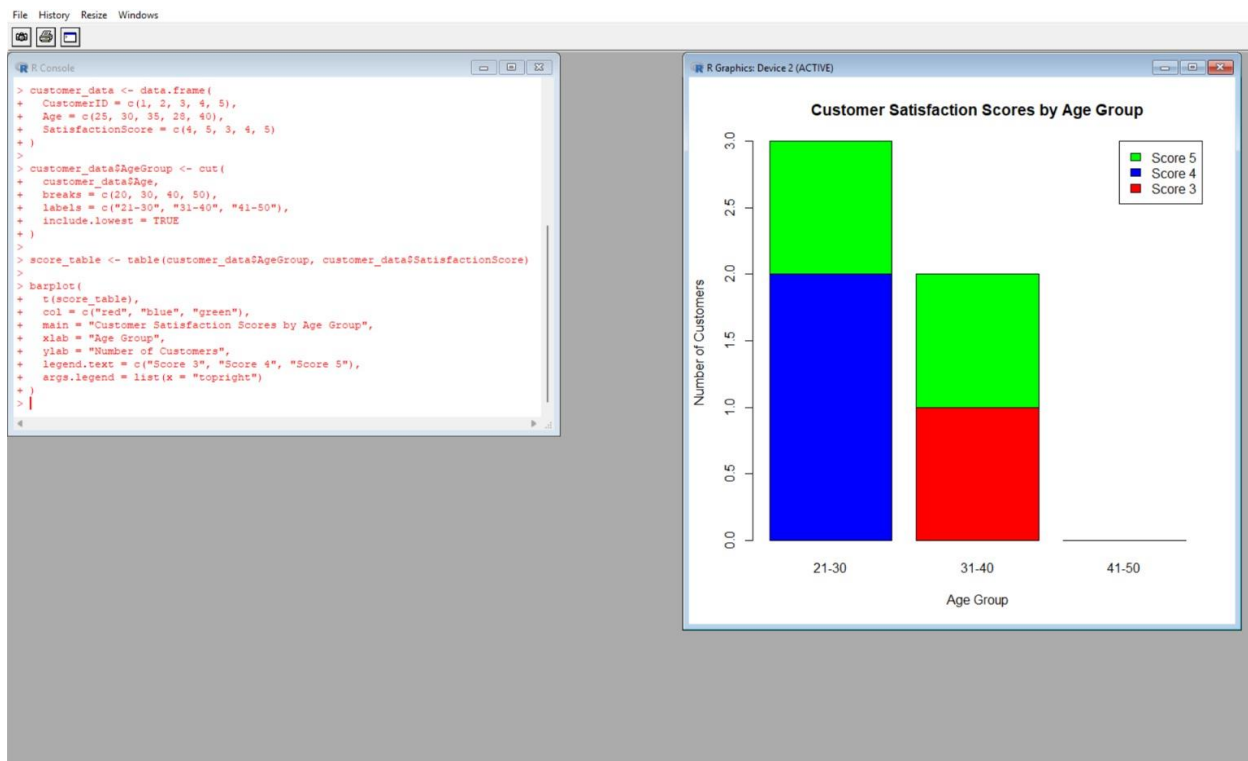
```
customer_data <- data.frame(  
  Customer_ID = c(1, 2, 3, 4, 5),  
  Age = c(25, 30, 35, 28, 40),  
  Satisfaction_Score = factor(c(4, 5, 3, 4, 5))  
)
```

```
customer_data$Age_Group <- cut(  
  customer_data$Age,  
  breaks = c(20, 30, 40, 50),  
  labels = c("21-30", "31-40", "41-50"),  
  right = TRUE  
)
```

```
ggplot(customer_data, aes(x = Age_Group, fill = Satisfaction_Score)) +  
  geom_bar() +
```

```
labs(
  title = "Customer Satisfaction Scores by Age Group",
  x = "Age Group",
  y = "Number of Customers",
  fill = "Satisfaction Score"
) +
theme_minimal()
```

OUTPUT :



4 :

CODE :

```
feedback <- c(
  "Good service",
  "Excellent support",
  "Fast delivery",
  "Friendly staff",
  "Good quality",
  "Excellent product",
  "Quick response",
  "Satisfied customer",
)
```

```
"Great experience",  
"Highly recommended"  
)
```

```
wordcloud(  
  words = feedback,  
  freq = rep(1, length(feedback)),  
  min.freq = 1,  
  random.order = FALSE,  
  colors = brewer.pal(8, "Dark2")  
)
```

OUTPUT :

Great experience
Excellent product
Friendly staff
Fast delivery
Good service
Excellent support
Good quality
Quick response
Satisfied customer

SET 3 : EMPLOYEE PERFORMANCE EVALUATION

1 :

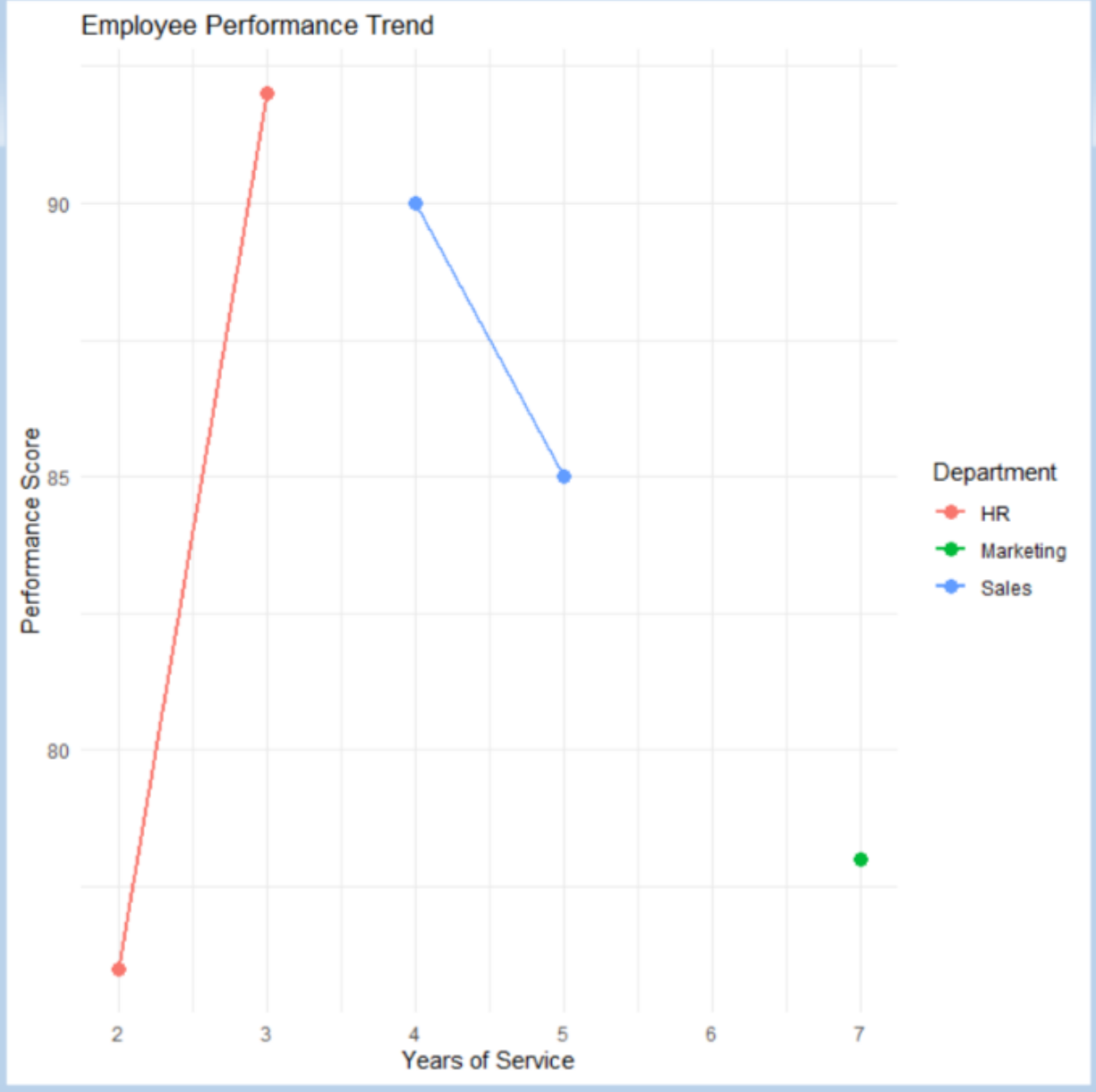
CODE :

```
library(ggplot2)
```

```
employee_data <- data.frame(  
  Employee_ID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  Years_of_Service = c(5, 3, 7, 4, 2),  
  Performance_Score = c(85, 92, 78, 90, 76)  
)
```

```
ggplot(employee_data, aes(x = Years_of_Service, y = Performance_Score, color =  
Department)) +  
  geom_line(linewidth = 1) +  
  geom_point(size = 3) +  
  labs(  
    title = "Employee Performance Trend",  
    x = "Years of Service",  
    y = "Performance Score",  
    color = "Department"  
  ) +  
  theme_minimal()
```

OUTPUT :



2 :

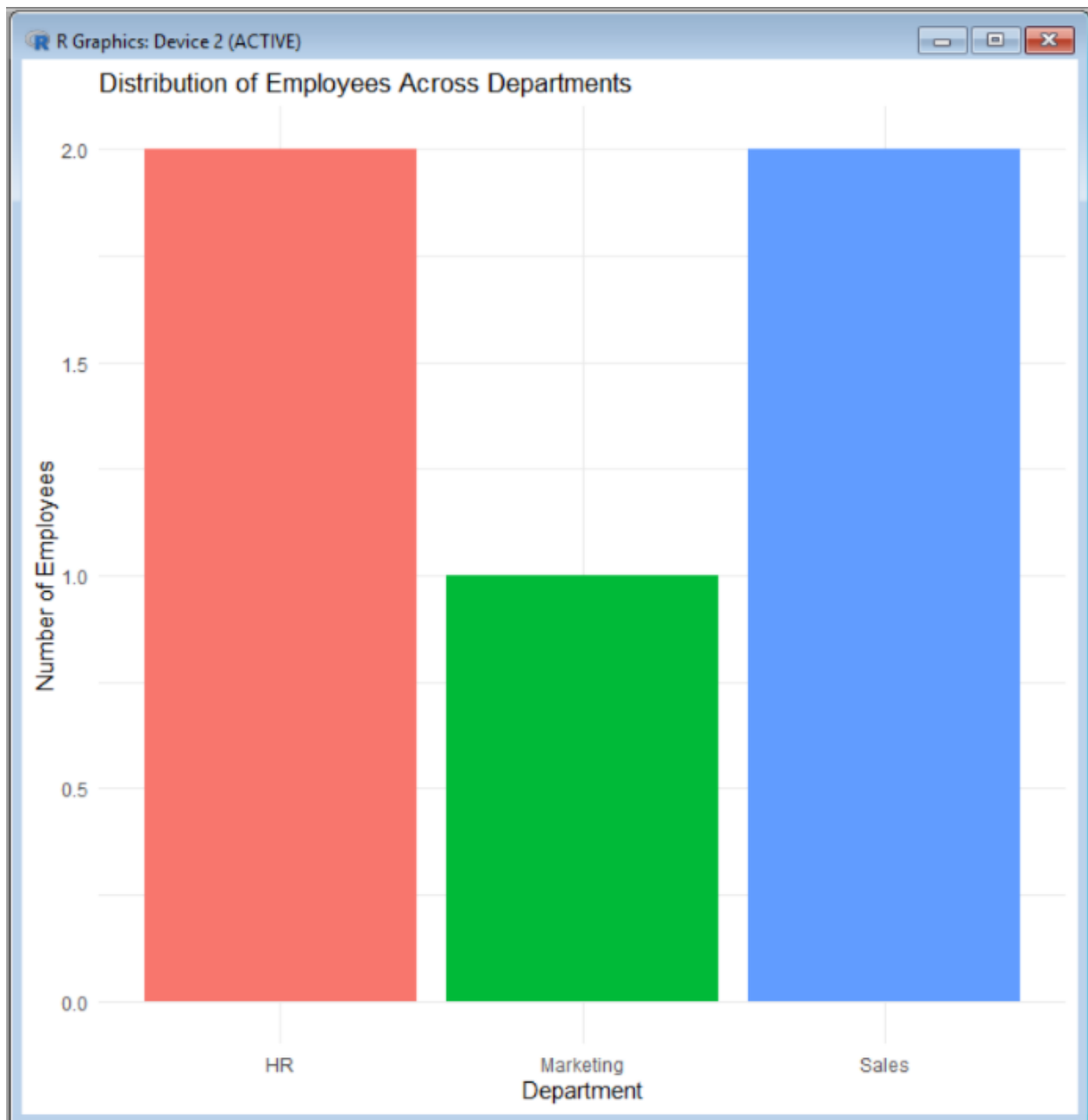
CODE :

```
library(ggplot2)

employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

ggplot(employee_data, aes(x = Department, fill = Department)) +
  geom_bar() +
  labs(
    title = "Distribution of Employees Across Departments",
    x = "Department",
    y = "Number of Employees"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT :



3 :

CODE :

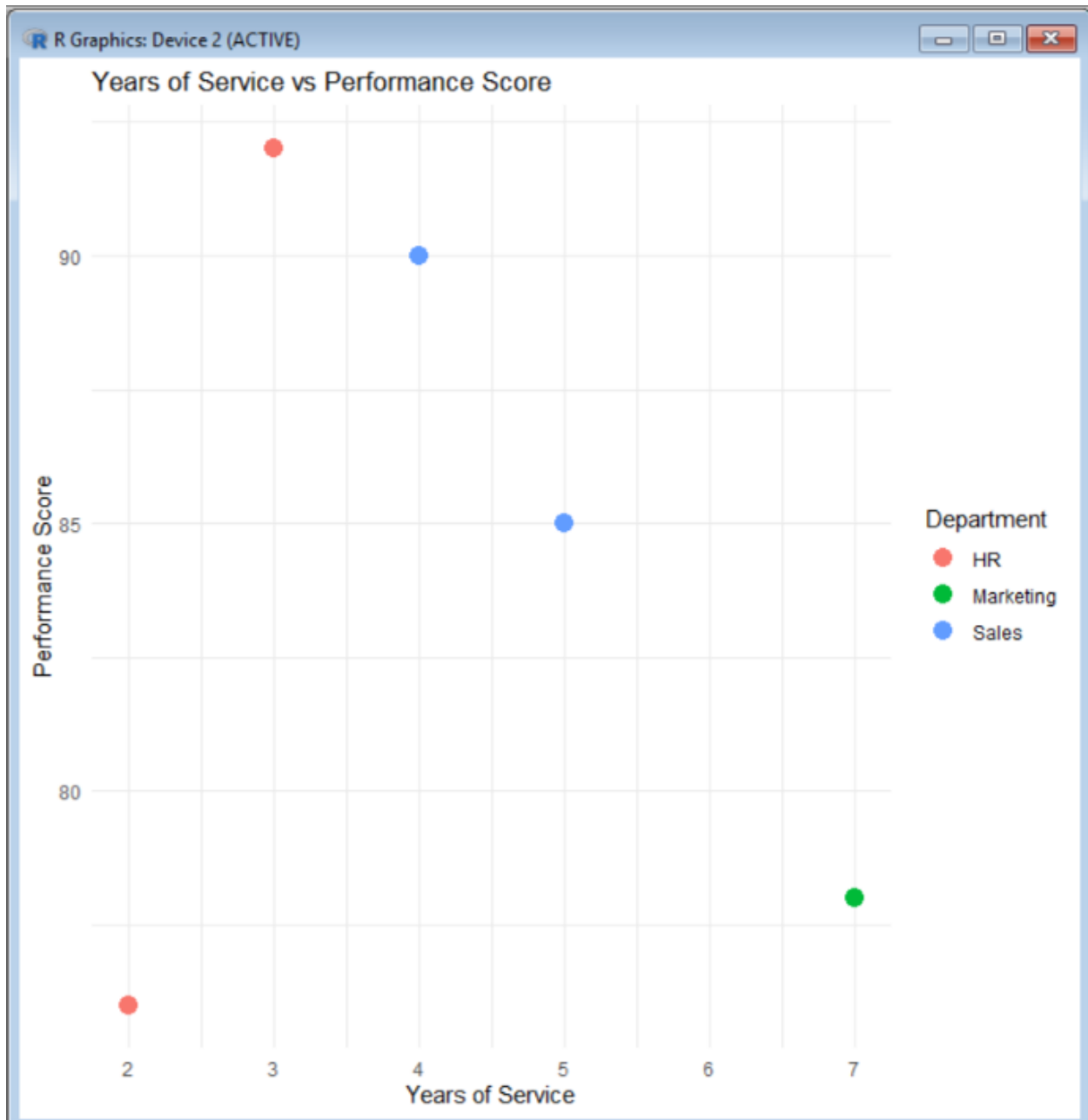
```
library(ggplot2)
```

```
employee_data <- data.frame(  
  Employee_ID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  Years_of_Service = c(5, 3, 7, 4, 2),  
  Performance_Score = c(85, 92, 78, 90, 76)
```

)

```
ggplot(employee_data, aes(x = Years_of_Service, y = Performance_Score, color =  
Department)) +  
  geom_point(size = 4) +  
  labs(  
    title = "Years of Service vs Performance Score",  
    x = "Years of Service",  
    y = "Performance Score"  
  ) +  
  theme_minimal()
```

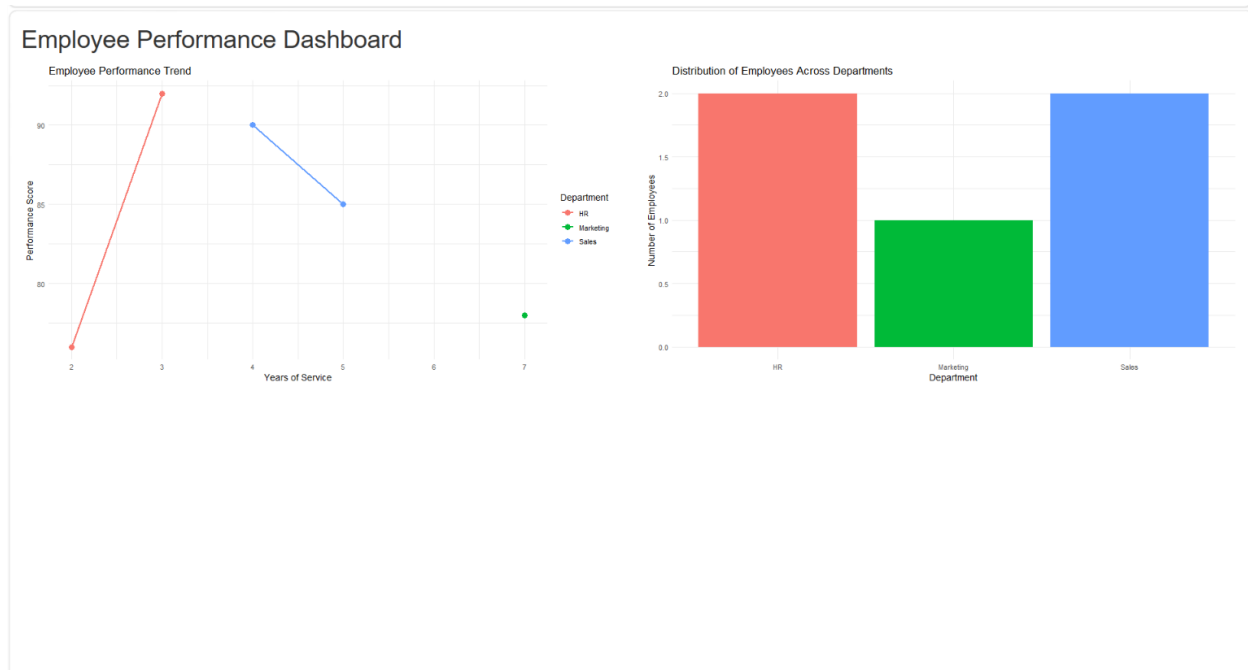
OUTPUT :



4 :

CODE :

OUTPUT :



SET 4 :

1.

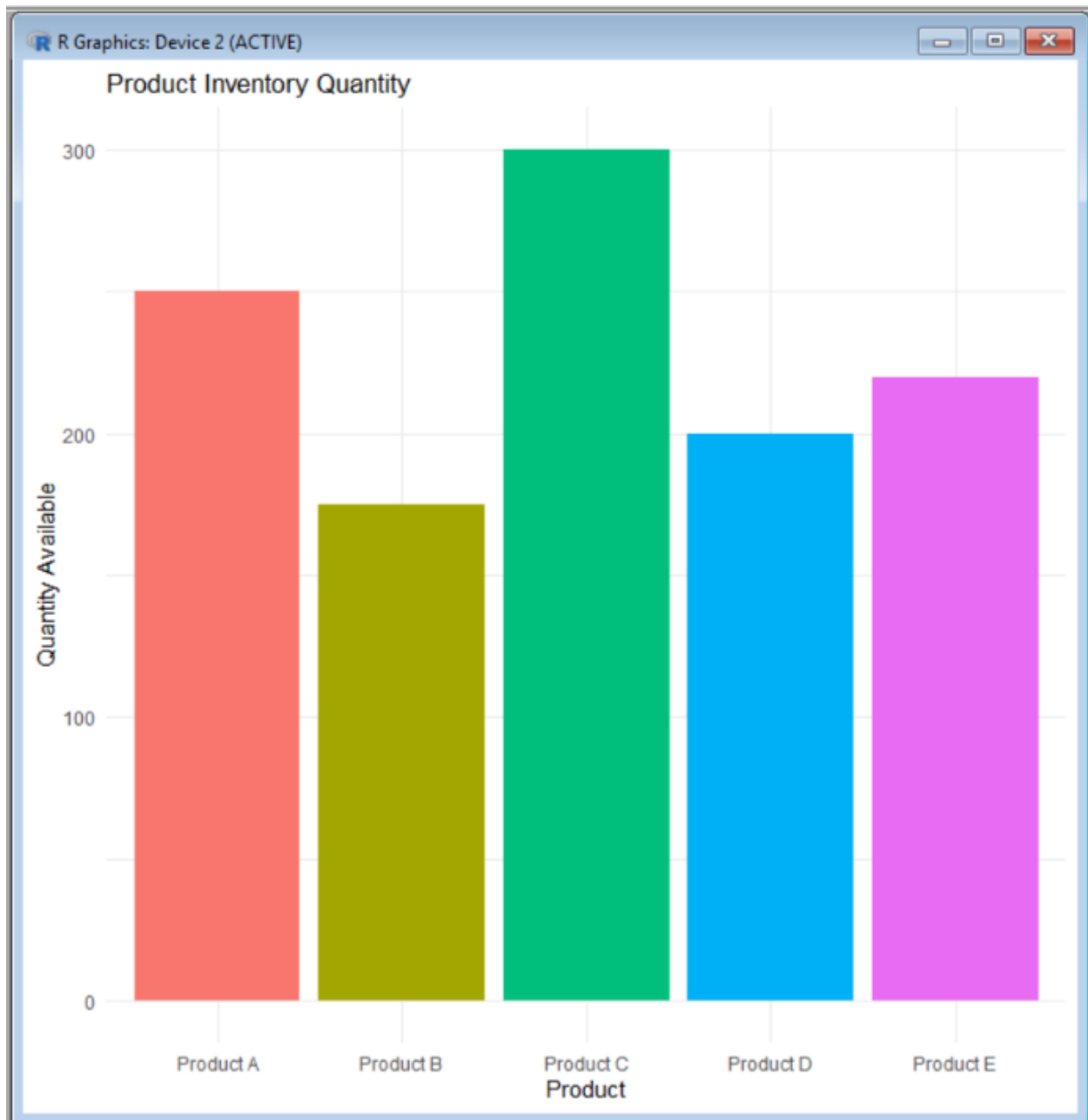
CODE :

```
library(ggplot2)
```

```
inventory_data <- data.frame(
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity = c(250, 175, 300, 200, 220)
)
```

```
ggplot(inventory_data, aes(x = Product, y = Quantity, fill = Product)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Product Inventory Quantity",
    x = "Product",
    y = "Quantity Available"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT :



2 :

CODE :

```
library(ggplot2)
```

```
inventory_data <- data.frame(  
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),  
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Accessories"),  
  Quantity = c(250, 175, 300, 200, 220)
```

)

```
ggplot(inventory_data, aes(x = Category, y = Quantity, fill = Product)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Inventory Quantity by Product Category",  
    x = "Category",  
    y = "Quantity Available",  
    fill = "Product"  
  ) +  
  theme_minimal()
```

OUTPUT :



3 :

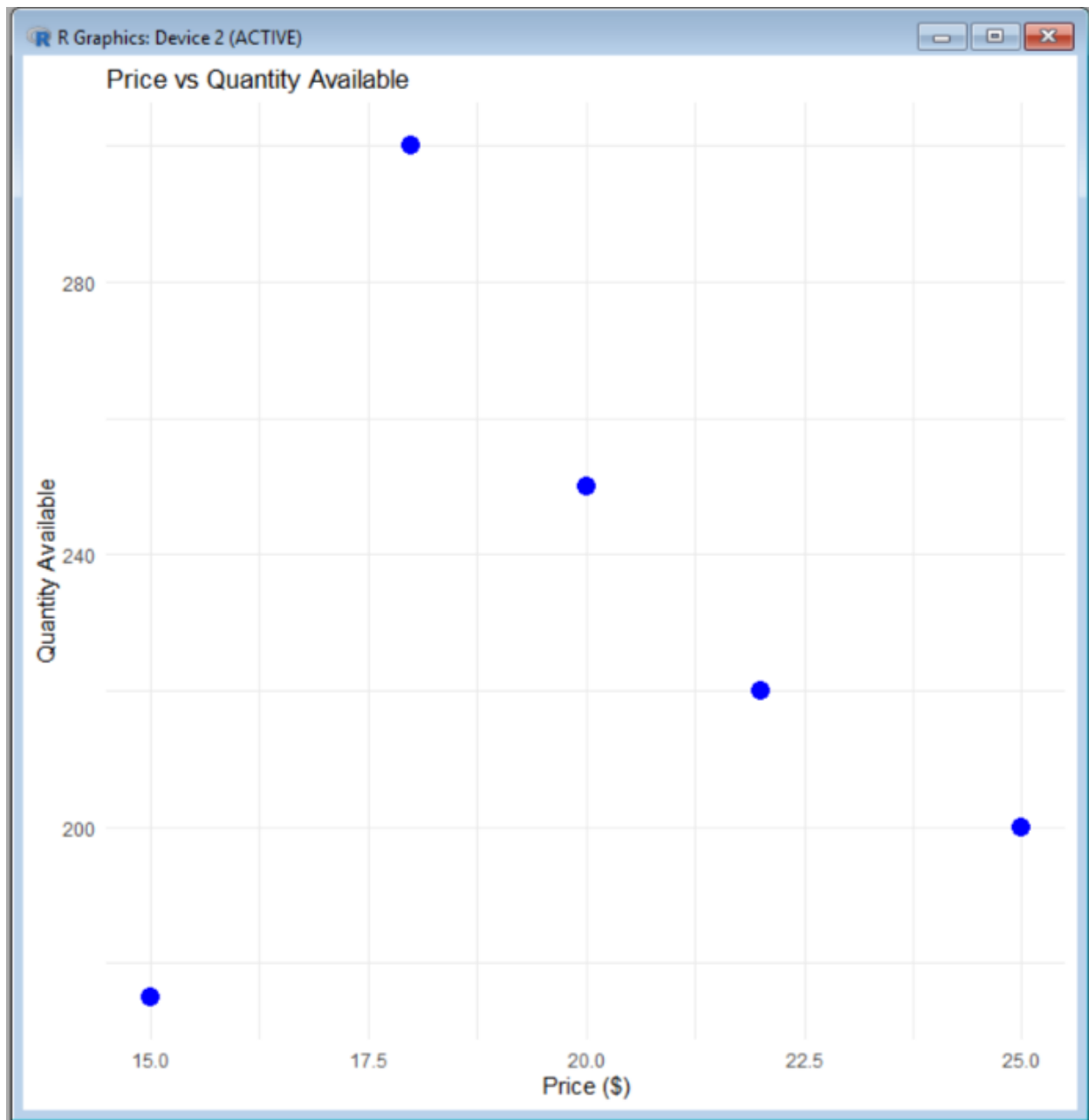
CODE :

```
library(ggplot2)
```

```
inventory_data <- data.frame(  
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),  
  Price = c(20, 15, 18, 25, 22),
```

```
Quantity = c(250, 175, 300, 200, 220)
)  
  
ggplot(inventory_data, aes(x = Price, y = Quantity)) +  
  geom_point(size = 4, color = "blue") +  
  labs(  
    title = "Price vs Quantity Available",  
    x = "Price ($)",  
    y = "Quantity Available"  
  ) +  
  theme_minimal()
```

OUTPUT :



4 :

CODE :

OUTPUT :



SET 5 : WEBSITE ANALYSIS

1 :

CODE :

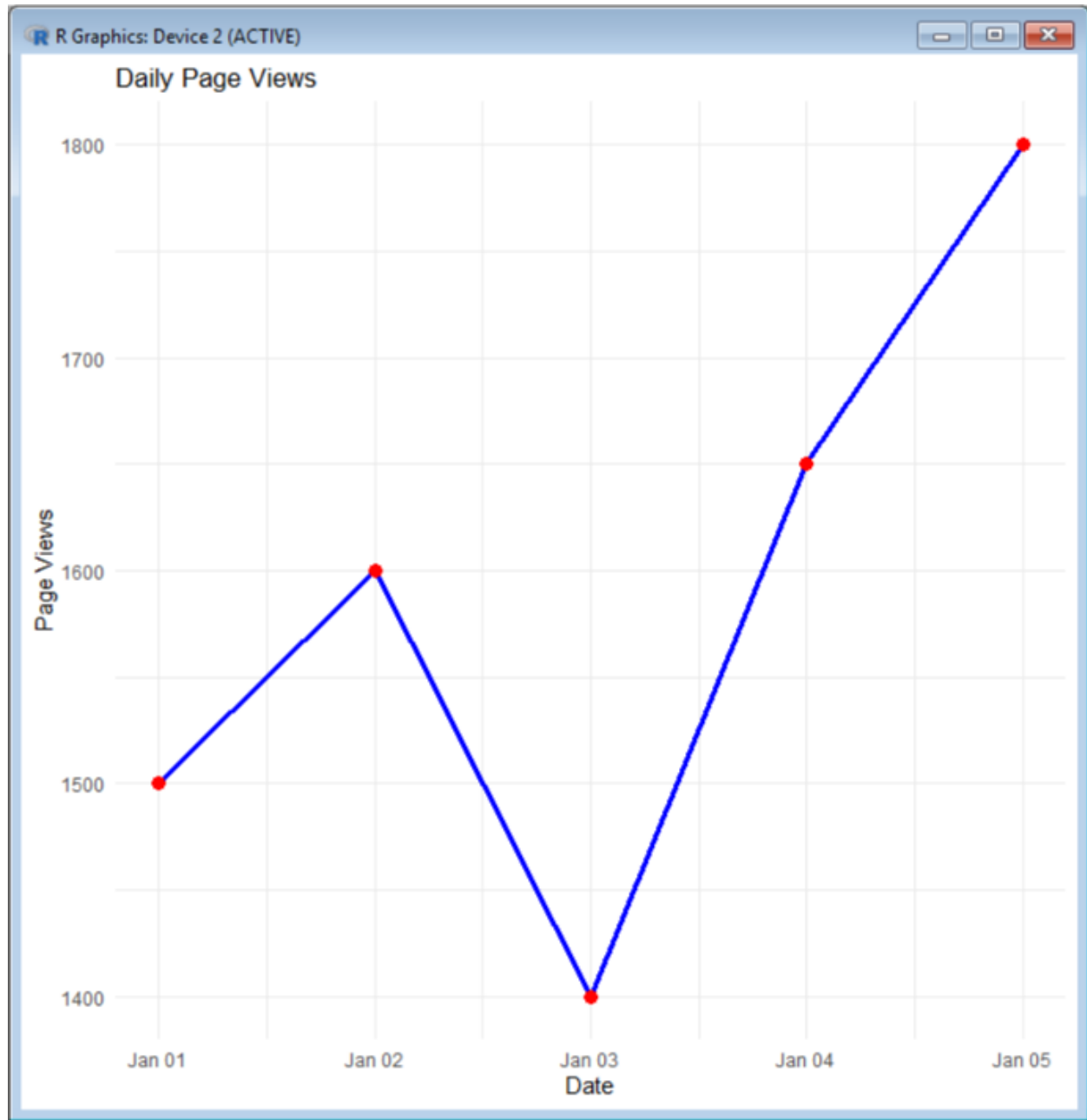
```
library(ggplot2)
```

```
website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)
)
```

```
ggplot(website_data, aes(x = Date, y = Page_Views, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Daily Page Views",
    x = "Date",
    y = "Page Views"
  ) +
```

```
theme_minimal()
```

OUTPUT :



2 :

CODE :

```
library(ggplot2)

website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

ggplot(website_data, aes(x = factor(Date), y = Click_Through_Rate, fill = factor(Date))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Click-Through Rate by Date",
    x = "Date",
    y = "Click-Through Rate (%)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

OUTPUT :



3 :

CODE :

```
library(ggplot2)
```

```
library(tidyr)
```

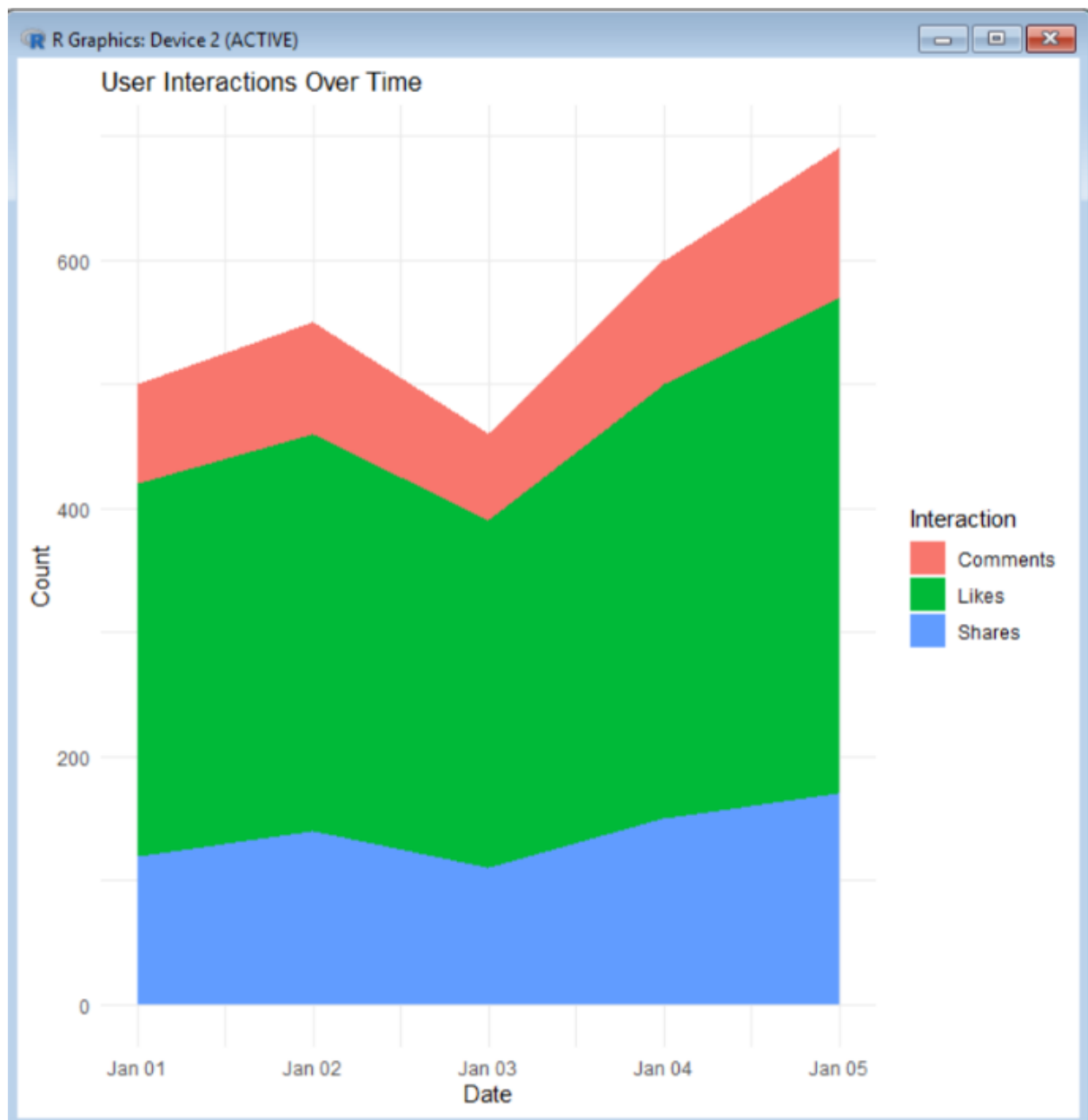
```
interaction_data <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),  
  Likes = c(300, 320, 280, 350, 400),  
  Shares = c(120, 140, 110, 150, 170),
```

```
Comments = c(80, 90, 70, 100, 120)
)

interaction_long <- pivot_longer(
  interaction_data,
  cols = c(Likes, Shares, Comments),
  names_to = "Interaction",
  values_to = "Count"
)

ggplot(interaction_long, aes(x = Date, y = Count, fill = Interaction)) +
  geom_area() +
  labs(
    title = "User Interactions Over Time",
    x = "Date",
    y = "Count"
  ) +
  theme_minimal()
```

OUTPUT :



4 :

CODE :

OUTPUT :

Website Traffic Dashboard

